AXOLABS METHOD VALIDATION REPORT FOR THE ANALYSIS OF 2439-EXNA IN RAT PLASMA	
Axolabs Project Number	AN403.0-24
Project Title	Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat plasma
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Start Date of Bioanalytical Test Phase	11-Sep-2024
End Date of Bioanalytical Test Phase	17-Sep-2024

SIGNATURE PAGE

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Alissa Coffey **SPONSOR'S REPRESENTATIVE**	Signed by: [redacted] Signer Name: Alissa Coffey Signing Reason: I approve this document Signing Time: 03-Dez-2024 | 15:33 CET 3E846AC45A5F4A42B5DCE31B3306D2F1

COMPLIANCE STATEMENT

This *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat plasma* was conducted in accordance with the procedures described herein.

The method validation was conducted in compliance with *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019) and following the rules in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

One minor deviation from the Method Validation Plan occurred during the method validation:

In section 9.10.4 of the method validation plan is described that the stability of the processed sample should be demonstrated for a minimum of 76 hours at 20 °C. To obtain this period during the method validation, a long-term equilibration sample is inserted into the sequence of the second validation batch prior to the QC set of the last processed sample stability time point. However, as the equilibration sample was set too short, only a process sample stability of 75.5 hours instead of 76 hours was validated. Since the difference is only 0.5 hours and the validated 75.5 hours is longer than an analysis batch with the maximum number of study samples can take, this deviation has no impact on the method validation results and is considered as minor. This deviation is documented in the Method Validation Report.

There were no deviations from respective SOPs.

The Method Validation Report represents an accurate and complete record of the results obtained during the method validation.

No deviations from the above regulations affected the overall integrity of this method validation.

Study Director:	Nicole Rottmann	Signiert von: [redacted] Name des Unterzeichners: Nicole Rottmann Signiergrund: Ich habe dieses Dokument verfasst Signierzeit: 03-Dez-2024 | 15:16 MEZ 87289BA16FC046139E8770AAAEB6DA46
	Name	Date and Signature

QUALITY ASSURANCE STATEMENT

This Method Validation Report titled *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in rat plasma* has been audited.

The reported results accurately reflect the raw data generated during this method validation.

As the bioanalytical testing is of short duration, routine and repetitive, Axolabs applies a process-based inspection program. The process-based inspection relevant to the activities performed during this method validation is listed below.

Table 1: Quality Assurance Inspections

Type(s) of Inspection(s)	Inspected Activities	Date(s) of the Inspection(s)	Date(s) of Reporting and Recipient(s)
Process-based (Report No.: II-24-003)	• Incoming Goods Check, storage, and labeling of study samples • Incoming Goods Check, storage, and labeling of reference standard, probe, blank matrix • Labeling and preparation of reagents/ solutions • Labelling and preparation of reference standard stock solution • Preparation of calibration standards and QC samples • Sample preparation • Lysate preparation and hybridization • HPLC setup and analysis • HPLC data processing and data evaluation • Disposal of study samples	03, 05, 11 & 18-Sep-2024	TSM: 11-Oct-2024 SD: 15-Oct-2024
Audit of Report and Data	Draft Report Version from 21-Nov-2024	27-Nov-2024	SD: 27-Nov-2024 TSM: 03-Dec-2024

TSM: Test Site Management. SD: Study Director.

In addition to the listed process-based inspection, facility-based inspections take place yearly.

Quality Assurance Axolabs: Anne Linhardt

Signiert von:
[redacted]
Name des Unterzeichners: Anne Linhardt
Signiergrund: Ich habe dieses Dokument geprüft
Signierzeit: 03-Dez-2024 | 15:26 MEZ
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Name | Date and Signature

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LIST OF ABBREVIATIONS

AB0X	Analysis Batch, AN403.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
BQL	Below Quantification Limit (Below the LLOQ)
CHMP	Committee for Human Medicinal Products
CoA	Certificate of Analysis
Conc.	Concentration
CV	Coefficient of Variation
DPBS	Dulbecco´s Phosphate Buffered Saline
DNA	Deoxyribonucleic Acid
EMA	European Medicines Agency
EU	European Union
FTS	Freeze-Thaw Stability
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
K_2-EDTA	Dipotassium Ethylenediaminetetraacetic Acid
LTS	Long-Term Stability
LLOQ	Lower Limit of Quantification
MDR	Method Development Report
n.a.	Not Applicable
OECD	Organization for Economic Co-operation and Development
PNA	Peptide Nucleic Acid
PrP	Prion Protein
PSS	Processed Sample Stability
QA	Quality Assurance
QC	Quality Control
RT	Room Temperature
SD	Standard Deviation
siRNA	small interfering Ribonucleic Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
STM	Standard Test Method
STS	Short-Term Stability
ULOQ	Upper Limit of Quantification
UV	Ultraviolet

1 SUMMARY

This report summarizes the results of the bioanalytical method validation for the analysis of 2439-exNA in rat plasma (Axolabs Project: AN403.0-24).

The assay is based on the specific hybridization of the antisense strands of 2439-exNA, a dimer composed of two identical siRNAs, with a complementary peptide nucleic acid (PNA) strand in a 1:1 stoichiometry relative to each antisense strand. The PNA is labeled with an Atto425 fluorescence dye and yields a specific signal in the subsequent analysis by Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) coupled to a fluorescence detector.

The method was developed without the need for any analyte extraction from rat plasma. Therefore, recovery of 100 % is maintained over the process (no internal standard is needed).

The bioanalytical method was previously developed at Axolabs GmbH and is described in the Axolabs Method Development Report (MDR) *MDR_2439-exNA_RatPlasma* (edition 01, effective date: 19-Aug-2024) that was used as the basis for this method validation (Supplemental File 1). During the method development phase, the suitability of the method for rat plasma was shown.

The method validation was carried out with the same conditions and parameters described in the MDR.

The final conditions of the validated method are described in the Standard Test Method (STM) *STM_Broad_2439-exNA_RatPlasma* (revision 01, effective date: 04-Oct-2024; Supplemental File 2).

The work was done according to the Method Validation Plan signed on 09-Sep-2024 by the Sponsor's Representative.

One minor deviation from the Method Validation Plan occurred during the method validation:

In section 9.10.4 of the method validation plan is described that the stability of the processed sample should be demonstrated for a minimum of 76 hours at 20 °C. To obtain this period during the method validation, a long-term equilibration sample is inserted into the sequence of the second validation batch prior to the QC set of the last processed sample stability time point. However, as the equilibration sample was set too short, only a process sample stability of 75.5 hours instead of 76 hours was validated. Since the difference is only 0.5 hours and the validated 75.5 hours is longer than an analysis batch with the maximum number of study samples can take, this deviation has no impact on the method validation results and is considered as minor. This deviation is documented in the Method Validation Report.

The results of the method validation are summarized in Table 2.

Table 2: Overall Summary Table

Items	Results	Section/Table	Comments
Analyte	2439-exNA	Section 3	n.a.
Reference Standard	2439-exNA	Section 4	n.a.
Methodology	PNA-AEX-HPLC-Assay with Fluorescence Detection	Section 8.1	n.a.
Biological Matrix (Anticoagulant)	rat plasma (K2-EDTA)	Section 5 Table 3	n.a.
Calibration Curve Range	1.0 ng/mL – 1,000.0 ng/mL	Section 9.2 Table 9	n.a.
Selectivity	5 times higher response (peak area) in the spiked QC-LLOQ samples compared to the signal in the blank samples	Section 9.3 Table 11	n.a.
Matrix Effect	No matrix effect observed for 7 (6 non-hemolyzed / non-lipemic and 1 hemolyzed matrix lot) individually tested matrix lots	Section 9.4 Table 12	n.a.
Carry-Over	<20 % of the response (peak area) of the LLOQ calibration standard.	Section 9.5 Table 13	n.a.
Between-Run Accuracy (for each QC concentration)	QC-LLOQ (1.0 ng/mL): 102.8 % QC-Low (3.0 ng/mL): 95.7 % QC-Mid (450.0 ng/mL): 98.1 % QC-High (900.0 ng/mL): 98.1 %	Section 9.7.2 Table 14	n.a.
Between-Run Precision (for each QC concentration)	QC-LLOQ (1.0 ng/mL): 6.5 % QC-Low (3.0 ng/mL): 3.7 % QC-Mid (450.0 ng/mL): 2.1 % QC-High (900.0 ng/mL): 2.3 %	Section 9.8.2 Table 14	n.a.
Dilution Integrity	Dilution QC: 500,000.0 ng/mL (Dilution factor: 1,000) Accuracy: 92.0 % Precision: 4.7 %	Section 9.9 Table 15	n.a.
Long-Term Stability	Ongoing (T=0) at -80 °C ± 20 °C	Section 9.10.1 Table 16	n.a.
Freeze-Thaw and Short-Term Stability	Demonstrated for 5 cycles at -80 °C ± 20 °C and for 4 hours on ice	Section 9.10.2 Table 17	n.a.
Bench-Top Stability (Robustness)	was not performed as test was covered by the short-term stability		
Processed Sample Stability	Demonstrated for 75.5 hours at 20 °C	Section 9.10.4 Table 18	n.a.
Short-Term Stability of Stock and Working Solution	Demonstrated for 4 hours at room temperature	Section 9.10.5.1 Table 19	n.a.

Items	Results	Section/Table	Comments
Long-Term Stability of Stock Solution	was not performed as data were provided by the Sponsor		

2 Compliance

The method validation was conducted in compliance with the following guidelines, regulations, or guidance documents:

- ICH guideline M10 on bioanalytical method validation and study sample analysis (EMA/CHMP/ICH/172948/2019); European Medicines Agency; 21 January 2023

Additionally, this method validation project was conducted following the rules as depicted in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

3 Analyte

The analyte 2439-exNA is a prion protein (PrP)-lowering divalent siRNA (Figure 1). The sense strands of the two identical siRNAs are linked to each other via a tetraethylene glycol linker.

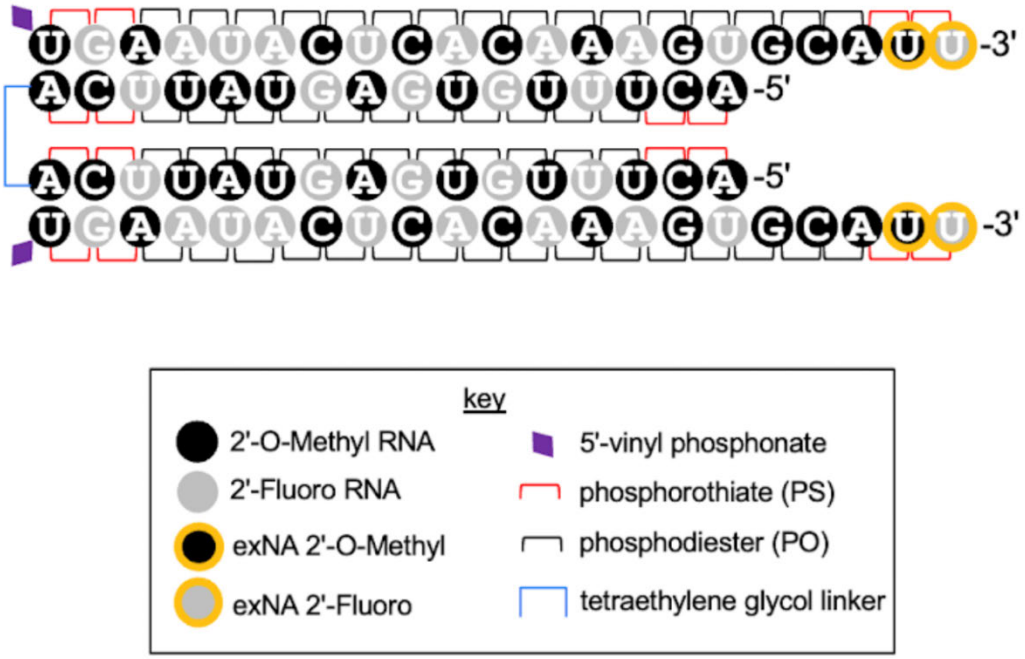


Figure 1: Structure of the PrP-lowering divalent siRNA 2439-exNA

4 Reference Standard

Identification: PrP-lowering divalent siRNA 2439-exNA (shortened to 2439-exNA in this document)

Structure:	Refer to Section 3
Physical Description:	Slightly yellow solution
Purity:	96.6 % (determined by non-denaturing HPLC)
Total Oligo Concentration:	9.9 mg/mL (determined by UV spectrophotometry)
Full-Length Concentration:	9.5634 mg/mL (corrected for purity stated on the CoA)
Supplier:	Argonaut Manufacturing Services (Carlsbad CA, USA)
Lot No.:	240621005
Manufacturing Date:	24-Jun-2024
Retest Date:	24-Jun-2026

With the exception of a document-based inspection and control of the corresponding Certificate of Analysis (CoA) provided by The Broad Institute, no further testing on identity or purity of the reference standard was conducted upon receipt.

As the reference standard was supplied by Argonaut Manufacturing Services as a liquid formulation and could be used directly for the preparation of the working solutions (using artificial plasma as diluent), separate aliquots were used for the calibration standards and the Quality Control (QC) samples. For this purpose, the reference standard was previously aliquoted into 100 µL aliquots (Eppendorf Protein LoBind tubes). After aliquoting, the reference standard was renamed to 2439-exNA_240621005_Stock1_04-Sep-2024 in accordance with internal nomenclature. For the preparation of the working solutions for the calibration standards and the Quality Control (QC) samples, the full-length concentration of the reference standard was used.

5 Blank Matrices

5.1 Sprague-Dawley Plasma

Blank matrices used in this method validation are listed in Table 3.

Table 3: Blank Matrices

Blank Matrix (Anticoagulant)	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Sprague-Dawley rat plasma (anticoagulated with K2-EDTA)	Male	30-Aug-2023	Envigo RMS B.V., Horst, Netherlands	HNL220301M	01-Mar-2027	AB01 – AB03
	Male			HNL220302M	02-Mar-2027	
	Male			HNL220921M	21-Sep-2027	
	Female			HNL221024F	24-Oct-2027	
	Female			HNL221025F	25-Oct-2027	
	Female			HNL230103F	03-Jan-2028	

Blank Matrix (Anticoagulant)	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Sprague-Dawley rat plasma (hemolyzed)	Female	13-Apr-2023	Biotrend Chemikalien GmbH, Cologne, Germany	7L01	23-Mar-2028	AB01

AB0X is the abbreviation for the analysis batch AN403.0-24_01_AB0X.

6 different blank rat plasma lots (non-hemolyzed and non-lipemic) were used for the evaluation of accuracy and precision, selectivity, and the matrix effect.

Additionally, one hemolyzed blank rat plasma lot was used for the evaluation of selectivity and the matrix effect.

A pool of different blank rat plasma lots was used for the preparation of calibration standards and QC samples.

5.2 Artificial Plasma

Type: Plasma

Receipt Date: 03-Sep-2024

Supplier: BioChemazone™, Leduc, Alberta, Canada

Lot No.: BZ273-0324E

Expiry Date: 30-Sep-2024

Analysis Batch: AB01 – AB03

Artificial Plasma was used for the preparation and the dilution of the working solutions.

6 PNA Probe

PNA Probe: PNA-BroadInst-01 from HLB Panagene Co., Ltd. (Daejeon, South Korea):

- N-terminus-Atto425 – OO – cdc ttt gtg dgt d – C-terminus, complementary to the antisense strands of 2439-exNA (OO: polyethylene glycol linker; d: 2,6-diaminopurine base modification)
  - Bulk powder was dissolved in 10 % acetonitrile (ACN) and 0.01 % Tween 20
  - Stock solution was diluted with 10 % ACN and 0.01 % Tween 20 to 2 µM (working solution)
- Lot No.: 241164
- Expiry Date: 22-Jul-2027

7 Reference Standard, PNA Probe, and Blank Matrices – Inventory and Disposition

Records of the receipt, distribution and storage of the reference standard, PNA probe, and blank matrices were maintained.

Unused reference standard will be retained for use in subsequent studies for the Sponsor.

Unused PNA probe and blank matrices will be retained for use in subsequent studies.

Expired reference standard will be disposed of unless the shelf life can be extended (new CoA required).

Expired PNA probe and blank matrices will be disposed of or will only be used for testing in non-regulated conditions.

8 Method

8.1 Method Summary

This report describes the validation of an AEX-HPLC method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA, a prion protein (PrP)-lowering divalent siRNA, in rat plasma samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent siRNA 2439-exNA.

The assay is a one-pot approach in which rat plasma lysates are prepared by Proteinase K treatment and the antisense strands of 2439-exNA is hybridized with the PNA in the same step. Subsequently, the samples are analyzed by AEX-HPLC coupled to a fluorescence detector.

In accordance with the MDR, the calibration standards and QC samples were diluted with artificial plasma and the samples were diluted with 4 M urea. Also, the injected sample volume was 50 µL, resulting in 2.5 µL plasma on column.

Quantification was performed based on an external calibration curve generated from a standard dilution series in rat plasma. Linear calibration curves (weighted 1/X) were calculated from 1.0 ng/mL to 1,000.0 ng/mL.

The detailed procedure is described in the STM, which is included as a supplemental file to this report.

8.2 Applicable Documents

- Axolabs Method Validation Plan for the analysis of 2439-exNA in rat plasma (Axolabs Project: AN403.0-24)

- Method Development Report (MDR) *MDR_2439-exNA_RatPlasma* (edition: 01, effective date: 19-Aug-2024)
- Standard Test Method (STM) *STM_Broad_2439-exNA_RatPlasma* (revision 01, effective date: 04-Oct-2024)
- KU-RGL-SOP-002, Revision 03 (formerly SOP REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*)
- KU-RGL-SOP-009, Revision 03 (formerly SOP REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*)
  - KU-RGL-INS-005, Revision 05 (formerly REG-109-I01) – PNA-HPLC-Assay SST

8.3 Computerized Systems

Critical computerized systems used in this method validation are listed in Table 4.

Table 4: Computerized Systems

System Name	Version / Revision	System Description
Thermo Fisher Ultimate 3000/Shimadzu LC-2050C HPLC System with a Shimadzu RF-20Axs Fluorescence Detector used with Chromeleon HPLC Software	7.2 SR5	HPLC Data collection and export to Excel
Microsoft Excel Spreadsheet: KU-RGL-ESS-002 (formerly: REG-102-S02) – Method Validation Data Report	03	Calculates results of calibration standard and QC sample accuracy and precision
Microsoft Excel Spreadsheet: KU-RGL-ESS-005 (formerly: REG-102-S05) – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot	01	Used as instruction and as the protocol for the assay execution for liquid matrices using the OnePot method: Calculates all sample preparation steps and the HPLC sequence for an analysis batch
Microsoft Excel Spreadsheet: KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions	02	Calculates all buffer preparation steps, including the preparation of the required stock solutions

8.4 HPLC Systems and Parameters

The HPLC systems used for the AEX-HPLC analyses are listed in Table 5.

Table 5: HPLC Systems

HPLC System	Module	Supplier	Model	Serial No.
HPLC-GLP-7	Degasser/Solvent Rack	Thermo Fisher	Ultimate SR-3000	7294808

(AB03)	Pump	Scientific	Ultimate LPG-3400SD	8135975
	Autosampler		Ultimate WPS-3000TSL	8135990
	Column Oven		Ultimate TCC-3000RS	6016238
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505573772
	Communication Unit		CBM-20A	L20235570188
HPLC System	**Module**	**Supplier**	**Model**	**Serial No.**
HPLC-GLP-S-18 (AB01 and AB02)	HPLC Kompakt System	Shimadzu	LC-2050C	L22975950036
	Fluorescence Detector		RF-20AXS	L20505974379

- Eluent Buffer A: 25 mM Trizma Hydrochloride solution, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea
- Eluent Buffer B: 25 mM Trizma Hydrochloride solution, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea, 1 M NaBr
- Eluent Buffer C: 10 % ACN, 4 M NaClO4, 2 M urea, 0.5x Triphosphate Buffer, adjusted to pH 2.5 with phosphoric acid
- Flow Rate: 1 mL/min
- Detection: Excitation @ 436 nm; Emission @ 484 nm; Gain: Medium 16
- Injection of 50 µL of the hybridized sample (corresponds to 2.5 µL matrix on column)
- Column Temperature: 45 °C
- Autosampler Temperature: 20 °C
- HPLC Program: Broad

Table 6: HPLC Gradient Table

Time [min]	% Eluent A	% Eluent B	% Eluent C	Curve
0.00	82	18	0	linear
1.00	82	18	0	linear
5.00	50	50	0	linear
5.10	0	0	100	linear
6.10	0	0	100	linear
6.20	82	18	0	linear
6.70	82	18	0	linear
6.80	0	0	100	linear
7.80	0	0	100	linear
7.90	82	18	0	linear
8.40	82	18	0	linear
8.50	0	0	100	linear
9.50	0	0	100	linear
9.60	82	18	0	linear
13.50	82	18	0	linear

8.5 HPLC Columns

Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010629; Lot Number 02306174 (used for AB01 - AB02)

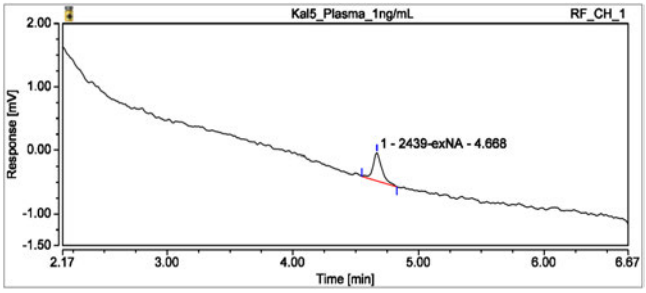
Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010628; Lot Number 02306174 (used for AB03)

8.6 Data Evaluation

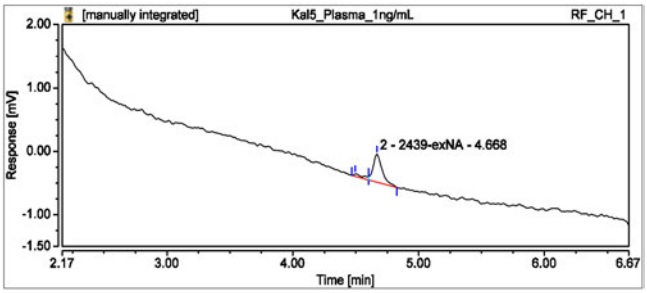
Data acquisition and integration were performed using the software package Chromeleon (Version 7.2) from Thermo Fisher Scientific.

For every injection, the peaks were integrated automatically by the HPLC chromatography software but are adjusted manually if the baseline and peak shoulders are not detected correctly by the HPLC software. When necessary, manual integration was performed by vertical peak splits at minima between two not fully resolved peaks. General rules for manual integration are documented in KU-RGL-SOP-009 (formerly REG-109).

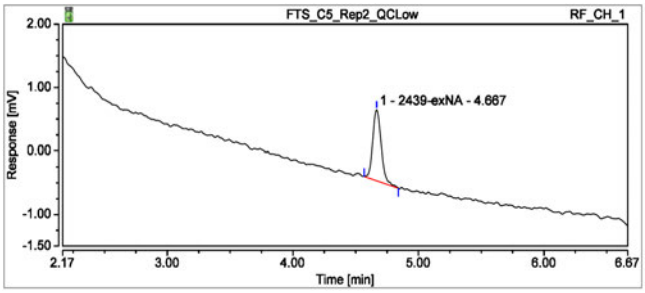
A report format of this file is generated containing values of retention time, peak area, and concentration of 2439-exNA in rat plasma and was saved in PDF format.



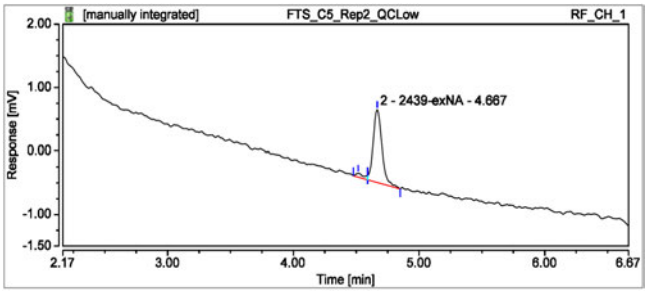
HPLC Chromatogram of Cal-level 2 – 1.0 ng/mL, Sample No. 4 in AB03 **(original)**



HPLC Chromatogram of Cal-level 2 – 1.0 ng/mL, Sample No. 4 in AB03 **(re-integrated)**



HPLC Chromatogram of FTS sample (5 Cycles; QC-Low), Sample No. 14 in AB03 **(original)**



HPLC Chromatogram of FTS sample (5 Cycles; QC-Low), Sample No 14 in AB03 **(re-integrated)**

Figure 2: Representative HPLC chromatograms (original and re-integrated)

8.7 Statistical Analysis

Mean values, standard deviations (SD), and coefficients of variation (CV) were calculated using validated Microsoft Excel spreadsheets.

9 Results and Discussion

9.1 System Suitability Test (SST)

The reproducibility of the HPLC system was determined by conducting an SST according to KU-RGL-INS-005 (formerly REG-109-I01) before measuring an analysis batch. Acceptance criteria are described in KU-RGL-INS-005 (formerly REG-109-I01), and the results of the SST run are attached to the documentation of each analysis batch.

A successful SST was completed before each reported analysis batch.

9.2 Calibration Curve

Calibration standards were generated by spiking known concentrations of 2439-exNA into blank rat plasma. The linear range was determined from 1.0 ng/mL to 1,000.0 ng/mL. The calibration levels used were 1.0 ng/mL, 5.0 ng/mL, 50.0 ng/mL, 100.0 ng/mL, 500.0 ng/mL, 1,000.0 ng/mL and one blank sample (rat plasma processed in the absence of 2439-exNA).

The calibration curves were prepared in duplicate for each analysis batch. Blank samples and calibration standards not meeting the acceptance criteria were not used to calculate the calibration curves.

In total, 6 calibration curves were analyzed in 3 different analysis batches.

9.2.1 Concentration-Response

The concentration-response relation was calculated by linear regression of the mean peak responses of the 6 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and coefficient of determination) from reported analysis batches are listed in Table 8.

9.2.2 Calibration Curve Accuracy

Acceptance Criteria:

- The back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).
- At least 75 % of the calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- 100 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 9.

Based on these results, the method can be used to determine concentrations of 2439-exNA in rat plasma over the concentration range of 1.0 ng/mL to 1,000.0 ng/mL.

9.2.3 Sensitivity/Lower Limit of Quantification (LLOQ)

The sensitivity of the assay is defined as the lowest 2439-exNA concentration that can be quantified with acceptable accuracy and precision – i.e., the LLOQ.

To confirm the LLOQ, 6 individually prepared samples with a nominal concentration at the LLOQ were analyzed (LLOQ calibration standards).

Acceptance Criteria:

- The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards should be within ± 20 % of the nominal concentration and the coefficient of variation (CV) should not exceed 20 %.

Results:

- No signal at the retention time of 2439-exNA was detected in the blank samples of the calibration curves and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards was 105.0 % of the nominal concentration with a CV of 5.2 %.

Data are listed in Table 9 and Table 10.

The LLOQ for 2439-exNA was confirmed at 1.0 ng/mL in rat plasma.

9.2.4 Upper Limit of Quantification (ULOQ)

The highest calibration standard in the calibration curve defines the ULOQ.

Six (6) individually prepared samples with a nominal concentration at the ULOQ were analyzed (ULOQ calibration standards).

Acceptance Criteria:

The mean back-calculated concentration of the 6 ULOQ calibration standards should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

The mean back-calculated concentration of the 6 ULOQ calibration standards was 100.0 % of the nominal concentration with a CV of 1.2 %.

Data are listed in Table 9.

The ULOQ for 2439-exNA was confirmed at 1,000.0 ng/mL in rat plasma.

9.3 Selectivity

The selectivity of the method was evaluated by comparison of unspiked lysate samples of 7 different blank rat plasma lots (6 non-hemolyzed and non-lipemic matrix lots, 1 hemolyzed matrix lot) with the same rat plasma samples spiked with 2439-exNA at the concentration of the LLOQ (QC-LLOQ).

Acceptance Criteria:

- The analyte response (peak area) in each QC-LLOQ sample should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same matrix lot.
- The back-calculated concentration each QC-LLOQ sample should be within ± 20 % of the nominal concentration.

Results:

- No signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 80.0 % – 110.0 % of the nominal concentration.

Data are listed in Table 11.

Selectivity of the method was confirmed for rat plasma.

9.4 Matrix Effect

The matrix effect was evaluated by analyzing triplicates of QC-Low and QC-High samples, each prepared using 7 different blank rat plasma lots (6 non-hemolyzed and non-lipemic matrix lots, 1 hemolyzed matrix lot).

Acceptance Criteria:

For each evaluated blank rat plasma lot, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations and the CV should not exceed 15 %.

Results:

- For 6 non-hemolyzed/non-lipemic matrix lots (HNL220301M, HNL220302M, HNL220921M, HNL221024F, HNL221025F, HNL230103F), the mean back-calculated concentrations of the QC samples ranged from 91.1 % – 100.0 % (QC-Low) and 95.2 % – 100.4 % (QC-High) of

the nominal concentrations, and CV values ranged from 0.0 % – 3.3 % (QC-Low) and 0.2 % – 1.3 % (QC-High).

- For the hemolyzed matrix lot (7L01), the mean back-calculated concentrations of the QC samples were 95.6 % (QC-Low) and 93.0 % (QC-High) of the nominal concentrations, and CV values were 5.3 % % (QC-Low) and 5.0 % % (QC-High).

Data are listed in Table 12.

Based on these results, there is no matrix effect for the assay regarding the analysis of 2439-exNA in rat plasma.

9.5 Carry-Over

Blank samples were injected after the ULOQ calibration standards to assess carry-over.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- No signal for 2439-exNA was detected in the carry-over blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 13.

No carry-over was observed for 2439-exNA in rat plasma. Therefore, no special precautions after the analysis of samples with a high concentration of 2439-exNA have to be implemented.

9.6 Quality Control (QC) Samples

Analysis of assay accuracy and precision, dilution integrity, and stability was conducted by analysis of QC samples of defined concentrations.

QC samples were prepared by spiking known amounts of 2439-exNA into rat plasma.

QC samples were prepared at 4 concentration levels which cover the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- within 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at 30 – 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL

- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

9.7 Accuracy

The accuracy of the analytical method describes the closeness of the determined concentration obtained by the method to the nominal concentration of 2439-exNA (expressed in percentage).

The analysis of assay accuracy was conducted using the QC samples at the 4 concentration levels that were measured as 6 individually prepared replicates per analytical run.

The analysis of assay accuracy was divided into within-run and between-run accuracy. The within-run accuracy was repeated 3 times, prepared by 2 technicians on different days, and analyzed on 2 independent HPLC systems. Thus, 3 independent data sets for the between-run accuracy were generated.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.7.1 Within-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentration of the QC samples at each of the 4 concentration levels and in each of the 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 98.3 % – 108.3 % (QC-LLOQ), 92.2 % – 99.4 % (QC-Low), 95.4 % – 99.8 % (QC-Mid) and 95.3 % – 100.0 % (QC-High) of the nominal concentrations.

Data are listed in Table 14.

9.7.2 Between-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentrations of all QC samples at each of the 4 concentration levels of all 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 102.8 % (QC-LLOQ), 95.7 % (QC-Low), 98.1 % (QC-Mid), and 98.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 14.

Based on these results, the assay for 2439-exNA in rat plasma is considered accurate.

9.8 Precision

The precision of the analytical method describes the closeness of repeated individual measures of the 2439-exNA concentration, expressed as coefficient of variation (CV).

The analysis of assay precision was conducted using the QC samples at the 4 concentration levels. All concentration levels were measured as 6 individually prepared replicates per analytical run.

As for accuracy, precision was demonstrated within a single analytical run and between different analytical runs. The same analytical runs and data were used for the demonstration of accuracy.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.8.1 Within-Run Precision

Acceptance Criteria:

The within-run CV of the QC samples at each of the 4 concentrations levels and in each of the 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 3.8 % – 7.4 % (QC-LLOQ), 1.8 % – 2.5 % (QC-Low), 0.5 % – 1.0 % (QC-Mid) and 0.7 % - 0.9 % (QC-High).

Data are listed in Table 14.

9.8.2 Between-Run Precision

Between-run precision was determined with an identical setup as for within-run precision but included the assessment of QC samples from 3 independent data sets prepared by 2 technicians on 3 different days on 2 independent HPLC systems.

Acceptance Criteria:

The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 6.5 % (QC-LLOQ), 3.7 % (QC-Low), 2.1 % (QC-Mid), and 2.3 % (QC-High).

Data are listed in Table 14.

Based on these results, the assay for 2439-exNA in rat plasma is considered precise.

9.9 Dilution Integrity

Dilution integrity was evaluated using QC samples prepared at the concentration level 500-fold (500,000.0 ng/mL) above the ULOQ (6 replicates for each dilution factor).

The 500,000.0 ng/mL QC samples (QC-500xULOQ) were analyzed and quantified after a 1:1,000 dilution.

The QC samples were diluted with 4 M urea.

Acceptance Criteria

For the evaluated dilution factor and concentration, the mean back-calculated concentration of the QC samples should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration was 92.0 % for the 500,000.0 ng/mL QC samples (1:1,000 dilution)
- The CV was 4.7 % for the 500,000.0 ng/mL QC samples (1:1,000)

Data are listed in Table 15.

Dilution of samples up to 1:1,000 up to a concentration of 500,000.0 ng/mL met the acceptance criteria for precision and accuracy. As the 1:1,000 dilution integrity testing fulfilled the acceptance criteria, no additional dilutions (e.g. 1:500) were tested.

9.10 Stability

The stability of 2439-exNA was analyzed under several conditions relevant to the assay setup.

2439-exNA was spiked at 3 concentration levels into rat plasma samples and analyzed in triplicate. The concentration levels were identical to the QC-Low, QC-High and QC-500xULOQ samples, with concentrations of 3.0 ng/mL (QC-Low), 900.0 ng/mL (QC-High) and 500,000.0 ng/mL (QC-500xULOQ). The QC-500xULOQ samples were evaluated after 1:1,000 dilution (using 4 M urea) to mimic potential samples above the validated ULOQ. Calibration curves were freshly prepared for each analytical run.

Long-term stability, freeze-thaw stability, and short-term stability were evaluated in polypropylene tubes (Eppendorf Protein LoBind tubes).

9.10.1 Long-Term Stability

Aliquots of 2439-exNA samples in rat plasma at the 3 QC concentration levels were stored at -80 °C ± 20 °C, and were and will be analyzed at up to 3 time points (T= 0, 3, and 12 months) in triplicate.

Acceptance Criteria:

For each evaluated condition, the mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

At the time the Method Validation Report was prepared, analyses of the first time point (T= 0) was available.

- At T= 0, the mean back-calculated concentrations were 101.1 % (QC-Low), 94.4 % (QC-High) and 92.6 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 16.

Long-term stability of 2439-exNA in rat plasma will be further monitored for up to 12 months. The final long-term stability data set will be added as an addendum to the Method Validation Report at the end of the study.

9.10.2 Freeze-Thaw and Short-Term Stability

Freeze-thaw and short-term stability of 2439-exNA in rat plasma were evaluated in a combined test.

Aliquots of 2439-exNA samples in rat plasma at the 3 QC concentration levels were stored at -80 °C ± 20 °C for at least 12 hours.

Thawing of the QC samples from -80 °C ± 20 °C was done on ice. After thawing, the QC samples were stored on ice for 4 hours before further processing, covering short-term stability requirements.

Subsequently, the QC samples were refrozen for at least 12 hours at -80 °C ± 20 °C to start the next freeze-thaw cycle and to repeat the process described above.

After 5 freeze-thaw cycles, the QC samples were processed and analyzed in triplicate according to the assay conditions.

Acceptance Criteria:

The mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After 5 freeze/thaw cycles with 4 hours on ice, the mean back-calculated concentrations were 87.8 % (QC-Low), 92.6 % (QC-High) and 87.7 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 17.

At the evaluated concentrations, 2439-exNA is stable in rat plasma when thawed up to 5 times on ice during sample analysis and for up to 4 hours during sample storage on ice.

9.10.3 Bench-Top Stability/Robustness

Since the assay is set up as a one-pot method that does not have separate lysis and hybridization steps, the bench-top stability was covered by the short-term stability (see section 9.10.2).

9.10.4 Stability of the Processed Sample/On-Instrument Stability (Autosampler)

Stability of the processed sample of 2439-exNA in rat plasma was assessed with 4 QC sets of QC-Low and QC-High triplicates that were stored on the HPLC autosampler at 20 °C and were analyzed at different time points, covering 75.5 hours between the analysis of the first and the last QC set.

Acceptance Criteria:

- For each evaluated time point, the mean back-calculated concentration of the QC-Low and QC-High samples should be within ± 15 % of the nominal concentrations.

- The QC set of last time point, which fulfills the acceptance criteria, defines processed sample stability.

Results:

- After time point 1, the mean back-calculated concentrations were 97.8 % (QC-Low) and 99.4 % (QC-High) of the nominal concentrations.
- After time point 2, the mean back-calculated concentrations were 97.8 % (QC-Low) and 99.4 % (QC-High) of the nominal concentrations.
- After time point 3, the mean back-calculated concentrations were 103.3 % (QC-Low) and 107.9 % (QC-High) of the nominal concentrations.
- After time point 4, the mean back-calculated concentrations were 103.3 % (QC-Low) and 108.2 % (QC-High) of the nominal concentrations.

Data are listed in Table 18.

At the evaluated concentrations, 2439-exNA in processed rat plasma samples is stable during storage on the HPLC autosampler at 20 °C for up to 75.5 hours.

If a longer period of sample storage on the autosampler should be necessary, the last QC sample set and calibration curve will verify the validity of the analysis.

9.10.5 Stability of Stock Solution and Working Solutions

9.10.5.1 Short-Term Stability of Stock Solution and Working Solutions

Aliquots of the reference standard stock solution of 2439-exNA with a concentration of approximately 9.5634 mg/mL in 1x DPBS (total oligonucleotide concentration corrected for the purity stated in the CoA), and the highest and the lowest concentration of the working solutions (working solution LLOQ/ULOQ) of 2439-exNA were stored at room temperature for 4 hours and analyzed in triplicate. The stock solution of the reference standard was diluted to QC-High concentration before analysis.

The samples were analyzed by comparing the instrument's response with a freshly thawed reference standard stock solution and freshly prepared working solutions of the same concentrations.

Acceptance Criteria:

For each evaluated condition, the mean back-calculated concentration of the QC-High samples (reference standard stock solution) and the working solution samples (each triplicates) should be within ± 15 % of the nominal concentrations (± 20 % for the working solution LLOQ).

Results:

After 4 hours at room temperature, the mean back-calculated concentrations were 83.3 % (working solution LLOQ), 95.9 % (working solution ULOQ), and 101.9 % (stock solution) of the nominal concentrations.

Data are listed in Table 19.

At the evaluated concentrations, 2439-exNA stock and working solutions covering the complete assay range are stable during storage for up to 4 hours at room temperature.

9.10.5.2 Long-Term Stability of Stock Solution

Long-term stability of the reference standard stock solution was not performed, as stability data will be provided by the Sponsor.

10 Deviations

During the bioanalytical method validation for the analysis of 2439-exNA in rat plasma, one minor deviation from the Method Validation Plan occurred:

In section 9.10.4 of the Method Validation Plan is described that the stability of the processed sample should be demonstrated for a minimum of 76 hours at 20 °C. To obtain this period during the method validation, a long-term equilibration sample is inserted into the sequence of the second validation batch prior to the QC set of the last processed sample stability time point. However, as the equilibration sample was set too short, only a process sample stability of 75.5 hours instead of 76 hours was validated. Since the difference is only 0.5 hours and the validated 75.5 hours is longer than an analysis batch with the maximum number of study samples can take, this deviation has no impact on the method validation results and is considered as minor.

There were no deviations from respective SOPs.

11 Conclusions

The bioanalytical method for the analysis of 2439-exNA in rat plasma was successfully validated and is described in the STM *STM_Broad_2439-exNA_RatPlasma* (Revision 01, effective date: 04-Oct-2024). All validation tests fulfilled the acceptance criteria described in the Method Validation Plan.

The method can be used within a linear range from 1.0 ng/mL to 1,000.0 ng/mL of 2439-exNA in rat plasma, applying a 1/X weighted linear regression model.

2439-exNA is stable in rat plasma for up to 5 freeze/thaw cycles. The evaluation of long-term stability of 2439-exNA in rat plasma is still ongoing, and the results will be added as an

addendum to this Method Validation Report after the finalization of the long-term stability testing.

12 List of Records to be Retained

All original data and documents generated during this method validation will be archived at Axolabs GmbH or at an Axolabs-employed GLP-compliant archive facility under appropriate conditions in accordance with German and EU legislation, and KU-RGL-SOP-005 (formerly SOP REG-105).

The following records will be retained:

- Final signed Method Validation Plan
- Critical communication, as appropriate
- Raw data:
  - HPLC file(s) with HPLC chromatograms, audit trails, and HPLC program
  - Assay protocols
  - System Suitability Test reports
  - CoAs or equivalent of PNA probe, matrices and reference standard
  - Other raw data (if applicable)
- QC and QA records of document and data reviews/audits
- Final signed Method Validation Report and all addenda

A backup file of all analytical HPLC data, together with an electronic PDF report of all analytical runs, and all electronic files generated during the method validation (reports, Excel-Files) are stored in electronic form.

13 DATA TABLES

Table 7: Analytical Batches for the Analysis of 2439-exNA in rat plasma

Run ID	Date of Analysis	Description	Status	Comments
AN403.0-24_01_AB01	11-Sep-2024	1st main validation batch	passed	n.a.
AN403.0-24_01_AB02	13-Sep-2024	2nd main validation batch	passed	n.a.
AN403.0-24_01_AB03	17-Sep-2024	3rd main validation batch	passed	n.a.

Table 8: Calibration Curve Parameters for the Analysis of 2439-exNA in rat plasma

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AN403.0-24_01_AB01	11-Sep-2024	4-9, 87-92	0.02355	-0.00741	0.99989
AN403.0-24_01_AB02	13-Sep-2024	4-9, 68-73	0.02176	-0.00404	0.99994
AN403.0-24_01_AB03	17-Sep-2024	4-9, 53-58	0.03538	-0.00099	0.99992

Table 9: Results of the Calibration Standards for the Analysis of 2439-exNA in rat plasma

		Cal-Level-2 (LLOQ)		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7 (ULOQ)	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	4 - 9	1.1	110.0 %	4.8	96.0 %	47.5	95.0 %	95.8	95.8 %	500.9	100.2 %	995.8	99.6 %
AB01	87 - 92	1.1	110.0 %	4.7	94.0 %	47.9	95.8 %	97.3	97.3 %	506.6	101.3 %	1,008.4	100.8 %
AB02	4 - 9	1.0	100.0 %	4.8	96.0 %	49.3	98.6 %	97.8	97.8 %	500.6	100.1 %	987.2	98.7 %
AB02	68 - 73	1.0	100.0 %	4.9	98.0 %	50.3	100.6 %	100.0	100.0 %	505.4	101.1 %	1,009.7	101.0 %
AB03	4 - 9	1.1	110.0 %	4.8	96.0 %	50.1	100.2 %	98.2	98.2 %	499.9	100.0 %	984.9	98.5 %
AB03	53 - 58	1.0	100.0 %	4.9	98.0 %	50.3	100.6 %	98.7	98.7 %	506.8	101.4 %	1,011.4	101.1 %
	Mean	1.05	105.0 %	4.82	96.3 %	49.23	98.5 %	97.97	98.0 %	503.37	100.7 %	999.57	100.0 %
	SD (±)	0.05		0.08		1.25		1.41		3.23		11.86	
	CV (%)	5.2 %		1.6 %		2.5 %		1.4 %		0.6 %		1.2 %	
	N	6		6		6		6		6		6	

Table 10: Sensitivity/Lower Limit of Quantification (LLOQ) for the Analysis of 2439-exNA in rat plasma

		Cal-Level-1 (Blank Sample)		Cal-Level-2 (LLOQ)	
		0.0 ng/mL		1.0 ng/mL	
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**
AB01	3, 4	0.0000	0.0 %	0.0186	1.1
AB01	86, 87	0.0000	0.0 %	0.0193	1.1
AB02	3, 4	0.0000	0.0 %	0.0184	1.0
AB02	67, 68	0.0000	0.0 %	0.0187	1.0
AB03	3, 4	0.0000	0.0 %	0.0366	1.1
AB03	52, 53	0.0000	0.0 %	0.0354	1.0

Table 11: Selectivity for the Analysis of 2439-exNA in rat plasma

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/mL		1.0 ng/mL		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**
AB01	11-Sep-2024	Male	HNL220301M	29, 37	0.0000	0.0 %	0.0185	1.1	110.0 %
AB01	11-Sep-2024	Male	HNL220302M	30, 38	0.0000	0.0 %	0.0175	1.1	110.0 %
AB01	11-Sep-2024	Male	HNL220921M	31, 39	0.0000	0.0 %	0.0185	1.1	110.0 %
AB01	11-Sep-2024	Female	HNL221024F	32, 40	0.0000	0.0 %	0.0190	1.1	110.0 %
AB01	11-Sep-2024	Female	HNL221025F	33, 41	0.0000	0.0 %	0.0188	1.1	110.0 %
AB01	11-Sep-2024	Female	HNL230103F	34, 42	0.0000	0.0 %	0.0171	1.0	100.0 %
AB01	11-Sep-2024	Female	7L01 (hemolyzed)	35, 43	0.0000	0.0 %	0.0121	0.8	80.0 %

Table 12: Matrix Effect for the Analysis of 2439-exNA in rat plasma

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	HNL220301M	45, 62	2.9	96.7 %	899.0	99.9 %
		46, 63	2.8	93.3 %	901.5	100.2 %
		47, 64	2.9	96.7 %	887.4	98.6 %
		Mean	2.87	95.6 %	895.97	99.6 %
		SD (±)	0.06		7.52	
		CV (%)	2.0 %		0.8 %	
		N	3		3	
AB01	HNL220302M	48, 65	2.8	93.3 %	889.9	98.9 %
		49, 66	2.9	96.7 %	892.6	99.2 %
		50, 67	2.9	96.7 %	881.0	97.9 %
		Mean	2.87	95.6 %	887.83	98.6 %
		SD (±)	0.06		6.07	
		CV (%)	2.0 %		0.7 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in rat plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB02	HNL220921M	26, 40	2.8	93.3 %	863.5	95.9 %
		27, 41	2.8	93.3 %	853.1	94.8 %
		28, 42	2.8	93.3 %	852.6	94.7 %
		Mean	2.80	93.3 %	856.40	95.2 %
		SD (±)	0.00		6.15	
		CV (%)	0.0 %		0.7 %	
		N	3		3	
AB02	HNL221024F	29, 43	2.7	90.0 %	866.3	96.3 %
		30, 44	2.8	93.3 %	851.4	94.6 %
		31, 45	2.7	90.0 %	858.7	95.4 %
		Mean	2.73	91.1 %	858.80	95.4 %
		SD (±)	0.06		7.45	
		CV (%)	2.1 %		0.9 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in rat plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB03	HNL221025F	30, 44	2.9	96.7 %	905.2	100.6 %
		31, 45	3.0	100.0 %	901.2	100.1 %
		32, 46	3.1	103.3 %	904.5	100.5 %
		Mean	3.00	100.0 %	903.63	100.4 %
		SD (±)	0.10		2.14	
		CV (%)	3.3 %		0.2 %	
		N	3		3	
AB03	HNL230103F	33, 47	3.0	100.0 %	898.0	99.8 %
		34, 48	3.0	100.0 %	885.1	98.3 %
		35, 49	2.9	96.7 %	908.5	100.9 %
		Mean	2.97	98.9 %	897.20	99.7 %
		SD (±)	0.06		11.72	
		CV (%)	1.9 %		1.3 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in rat plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	7L01 (hemolyzed)	51, 68	2.9	96.7 %	849.1	94.3 %
		52, 69	3.0	100.0 %	872.2	96.9 %
		53, 70	2.7	90.0 %	790.9	87.9 %
		Mean	2.87	95.6 %	837.40	93.0 %
		SD (±)	0.15		41.89	
		CV (%)	5.3 %		5.0 %	
		N	3		3	

Table 13: Carry-Over for the Analysis of 2439-exNA in rat plasma

		Cal-Level-2 (LLOQ)			Cal-Level-7 (ULOQ)			Blank Sample for Carry-Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Peak Area [mv*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**
AB01	4, 9, 10	0.0186	1.1	110.0 %	23.4419	995.8	99.6 %	0.0000	BQL	0.0 %
AB01	87, 92, 93	0.0193	1.1	110.0 %	23.7380	1,008.4	100.8 %	0.0000	BQL	0.0 %
AB02	4, 9, 10	0.0184	1.0	100.0 %	21.4773	987.2	98.7 %	0.0000	BQL	0.0 %
AB02	68, 73, 74	0.0187	1.0	100.0 %	21.9673	1,009.7	101.0 %	0.0000	BQL	0.0 %
AB03	4, 9, 10	0.0366	1.1	110.0 %	34.8469	984.9	98.5 %	0.0000	BQL	0.0 %
AB03	53, 58, 59	0.0354	1.0	100.0 %	35.7849	1,011.4	101.1 %	0.0000	BQL	0.0 %

Table 14: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in rat plasma

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB01	11-Sep-2024	JSt	HPLC-GLP-S-18	37, 45, 55, 62	1.1	110.0 %	2.9	96.7 %	448.6	99.7 %	899.0	99.9 %
				38, 46, 56, 63	1.1	110.0 %	2.8	93.3 %	442.8	98.4 %	901.5	100.2 %
				39, 47, 57, 64	1.1	110.0 %	2.9	96.7 %	446.3	99.2 %	887.4	98.6 %
				40, 48, 58, 65	1.1	110.0 %	2.8	93.3 %	444.3	98.7 %	889.9	98.9 %
				41, 49, 59, 66	1.1	110.0 %	2.9	96.7 %	444.4	98.8 %	892.6	99.2 %
				42, 50, 60, 67	1.0	100.0 %	2.9	96.7 %	446.8	99.3 %	881.0	97.9 %
Within-Run Accuracy/Precision I				**Mean**	1.08	108.3 %	2.87	95.6 %	445.53	99.0 %	891.90	99.1 %
				SD (±)	0.04		0.05		2.09		7.56	
				CV (%)	3.8 %		1.8 %		0.5 %		0.8 %	
				N	6		6		6		6	
AB02	13-Sep-2024	JSt	HPLC-GLP-S-18	19, 26, 33, 40	1.1	110.0 %	2.8	93.3 %	435.6	96.8 %	863.5	95.9 %
				20, 27, 34, 41	1.0	100.0 %	2.8	93.3 %	430.7	95.7 %	853.1	94.8 %
				21, 28, 35, 42	1.1	110.0 %	2.8	93.3 %	428.3	95.2 %	852.6	94.7 %
				22, 29, 36, 43	1.0	100.0 %	2.7	90.0 %	427.4	95.0 %	866.3	96.3 %
				23, 30, 37, 44	1.0	100.0 %	2.8	93.3 %	430.9	95.8 %	851.4	94.6 %
				24, 31, 38, 45	0.9	90.0 %	2.7	90.0 %	423.0	94.0 %	858.7	95.4 %
Within-Run Accuracy/Precision II				**Mean**	1.02	101.7 %	2.77	92.2 %	429.32	95.4 %	857.60	95.3 %
				SD (±)	0.08		0.05		4.21		6.25	
				CV (%)	7.4 %		1.9 %		1.0 %		0.7 %	
				N	6		6		6		6	

Table 14: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in rat plasma (Continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB03	17-Sep-2024	YE	HPLC-GLP-7	23, 30, 37, 44	1.0	100.0 %	2.9	96.7 %	451.7	100.4 %	905.2	100.6 %
				24, 31, 38, 45	1.0	100.0 %	3.0	100.0 %	450.3	100.1 %	901.2	100.1 %
				25, 32, 39, 46	0.9	90.0 %	3.1	103.3 %	452.0	100.4 %	904.5	100.5 %
				26, 33, 40, 47	1.0	100.0 %	3.0	100.0 %	443.2	98.5 %	898.0	99.8 %
				27, 34, 41, 48	1.0	100.0 %	3.0	100.0 %	448.0	99.6 %	885.1	98.3 %
				28, 35, 42, 49	1.0	100.0 %	2.9	96.7 %	449.3	99.8 %	908.5	100.9 %
Within-Run Accuracy/Precision III				**Mean**	0.98	98.3 %	2.98	99.4 %	449.08	99.8 %	900.42	100.0 %
				SD (±)	0.04		0.08		3.25		8.32	
				CV (%)	4.2 %		2.5 %		0.7 %		0.9 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	1.03	102.8 %	2.87	95.7 %	441.31	98.1 %	883.31	98.1 %
				SD (±)	0.07		0.11		9.38		20.28	
				CV (%)	6.5 %		3.7 %		2.1 %		2.3 %	
				N	18		18		18		18	

Table 15: Dilution Integrity for the Analysis of 2439-exNA in rat plasma

		Dilution 1:1,000	
		500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	72	453,719.0	90.7 %
	73	464,361.3	92.9 %
	74	475,108.1	95.0 %
	75	475,922.9	95.2 %
	76	419,600.5	83.9 %
	77	471,406.9	94.3 %
	Mean	460,019.78	92.0 %
	SD (±)	21,459.26	
	CV (%)	4.7 %	
	N	6	

Underlined QC sample did not fulfill the acceptance criterion but was included in the statistical calculations (Mean, SD and CV).

Table 16: Long-Term Stability for the Analysis of 2439-exNA in rat plasma

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	T= 0 (-80 °C ± 20 °C)	19, 22, 25	3.0	100.0 %	853.4	94.8 %	468,410.0	93.7 %
		20, 23, 26	3.0	100.0 %	853.2	94.8 %	465,500.5	93.1 %
		21, 24, 27	3.1	103.3 %	842.7	93.6 %	455,158.5	91.0 %
		Mean	3.03	101.1 %	849.77	94.4 %	463,023.00	92.6 %
		SD (±)	0.06		6.12		6,964.49	
		CV (%)	1.9 %		0.7 %		1.5 %	
		N	3		3		3	

Table 17: Freeze-Thaw and Short-Term Stability for the Analysis of 2439-exNA in rat plasma

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB03	• FTS: 5 cycles (-80 °C ± 20 °C) • STS: 4 hours on ice	13, 16, 19	2.7	90.0 %	830.8	92.3 %	438,521.6	87.7 %
		14, 17, 20	2.6	86.7 %	831.5	92.4 %	429,979.6	86.0 %
		15, 18, 21	2.6	86.7 %	837.4	93.0 %	447,630.4	89.5 %
		Mean	2.63	87.8 %	833.23	92.6 %	438,710.53	87.7 %
		SD (±)	0.06		3.63		8,826.92	
		CV (%)	2.2 %		0.4 %		2.0 %	
		N	3		3		3	

Table 18: Processed Sample Stability for the Analysis of 2439-exNA in rat plasma

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	11-Sep-2024 (Determination 1)	12, 15	3.0	100.0 %	894.6	99.4 %
		13, 16	2.9	96.7 %	900.1	100.0 %
		14, 17	2.9	96.7 %	889.6	98.8 %
		Mean	2.93	97.8 %	894.77	99.4 %
		SD (±)	0.06		5.25	
		CV (%)	2.0 %		0.6 %	
		N	3		3	
AB01	12-Sep-2024 (Determination 2)	79, 82	2.9	96.7 %	888.2	98.7 %
		80, 83	3.0	100.0 %	896.9	99.7 %
		81, 84	2.9	96.7 %	897.6	99.7 %
		Mean	2.93	97.8 %	894.23	99.4 %
		SD (±)	0.06		5.24	
		CV (%)	2.0 %		0.6 %	
		N	3		3	

Table 18: Processed Sample Stability for the Analysis of 2439-exNA in rat plasma (Continued)

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	13-Sep-2024 (Determination 3)	12, 15	3.1	103.3 %	975.9	108.4 %
		13, 16	3.1	103.3 %	970.5	107.8 %
		14, 17	3.1	103.3 %	967.9	107.5 %
		Mean	3.10	103.3 %	971.43	107.9 %
		SD (±)	0.00		4.08	
		CV (%)	0.0 %		0.4 %	
		N	3		3	
AB02	14-Sep-2024 (Determination 4)	60, 63	3.1	103.3 %	969.0	107.7 %
		61, 64	3.1	103.3 %	981.5	109.1 %
		62, 65	3.1	103.3 %	970.0	107.8 %
		Mean	3.10	103.3 %	973.50	108.2 %
		SD (±)	0.00		6.95	
		CV (%)	0.0 %		0.7 %	
		N	3		3	

Table 19: Short-Term Stability of Stock and Working Solutions for the Analysis of 2439-exNA in rat plasma

Run ID	Validation Parameters	Sample No.	Working Solution LLOQ		Working Solution ULOQ		Stock Solution (diluted to QC-High)	
			1.0 ng/mL		1,000.0 ng/mL		900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 20 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	4 h at RT	47, 50, 53	0.9	90.0 %	977.4	97.7 %	907.9	100.9 %
		48, 51, 54	0.8	80.0 %	967.6	96.8 %	923.5	102.6 %
		49, 52, 55	0.8	80.0 %	933.4	93.3 %	919.1	102.1 %
		Mean	0.83	83.3 %	959.47	95.9 %	916.83	101.9 %
		SD (±)	0.06		23.10		8.04	
		CV (%)	6.9 %		2.4 %		0.9 %	
		N	3		3		3	

14 SUPPLEMENTARY INFORMATION

14.1 **Supplemental File 1**

Axolabs *MDR_2439-exNA_RatPlasma*

14.2 Supplemental File 2

Axolabs *STM_Broad_2439-exNA_RatPlasma*